



Australian
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Basic Training

Build a bigram language model that tracks which words follow which other words in text.

You will need

- some text (e.g. a few pages from a kids book, but can be anything)
- pen, pencil and grid paper

Your goal

To produce a grid that captures the patterns in your input text data. This grid is your *bigram language model*. **Stretch goal:** keep training your model on more input text.

Key idea

Language models learn by counting patterns in text. “Training” means building/constructing a model (i.e. filling out the grid) to track which words follow other words.



Algorithm

1. preprocess your text:

- convert everything to lowercase
- treat words, commas and full stops as separate “words” (and ignore all other punctuation and whitespace)

2. set up your grid:

- take the first word from your text
- write it in both the first row header and first column header of your grid

3. fill in the grid one word pair at a time:

- find the row for the first word (in your training text) and the column for the second word
- add a tally mark in that cell (if the word isn’t in the grid yet, add a new row *and* column for it)
- shift along by one word (so the second word becomes your “first” word)
- repeat until you’ve gone through the entire text

Example

Original text: “See Spot run. See Spot jump. Run, Spot, run. Jump, Spot, jump.”

Preprocessed text: see spot run . see spot jump . run , spot , run . jump , spot , jump .

After the first two words (see spot) the model looks like:

	see	spot				
see						
spot						

After the full text the model looks like:

	see	spot	run	.	jump	,
see						
spot						
run						
.						
jump						
,						